

ENTHUSIAST, LEADER & ACHIEVER COURSE

ALL PHASE

TARGET : PRE-MEDICAL 2020

Test Type : **MAJOR**

Test Pattern : **NEET(UG)**

TEST DATE : 26 - 03 - 2020

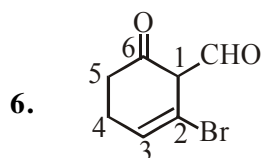
TEST SYLLABUS : 01, 02 & 03

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	1	3	3	4	1	4	1	1	4	4	2	4	2	2	3	3	1	1	2	4	3	4	2	3	1	3	2	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	4	4	1	2	2	1	3	1	1	3	3	3	1	3	4	4	3	2	3	2	1	2	3	4	3	2	4	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	1	2	4	2	4	4	4	1	3	2	4	2	4	2	1	3	2	3	2	2	4	2	4	4	3	4	4	3	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	4	3	3	4	3	3	1	4	4	1	1	1	4	3	4	4	1	2	3	4	3	3	4	4	3	1	4	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	3	3	3	1	3	3	3	2	3	2	3	3	2	3	4	1	3	4	4	3	1	1	3	4	1	4	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	1	4	4	4	3	2	1	1	2	4	2	4	1	1	3	3	1	4	3	1	4	3	1	3	3	1	3	1	2

HINT – SHEET

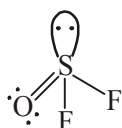
1.
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_2\text{C} = \text{CH} - \overset{*}{\text{C}} - \text{H}_2\text{C} - \text{CH}_3 \\ | \\ \text{H} \end{array}$$

3-methyl 1-pentene have one chiral carbon atom



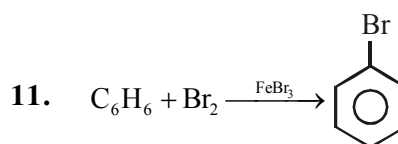
7.
$$\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$$

9. (1) Bond angle $\alpha = \beta = \gamma$
(3) In OSF_2 angle $\text{O}-\text{S}-\text{F} > \text{Angle F}-\text{S}-\text{F}$



10.
$$\text{A} \rightleftharpoons \text{D}$$

$$K = K_1 \times K_2 \times K_3$$



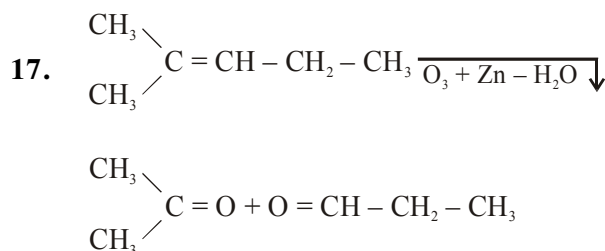
12. Group of electrophile given in 2nd is true.

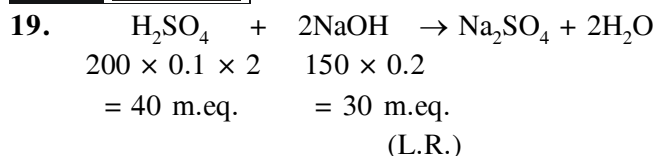
15. NaOH is a base
 $\text{pH} > 7$

$$\frac{10^{-8} \text{ M NaOH}}{8 > \text{pH} > 7}$$

16. Stability of free radical $\propto +M, +H, +I$

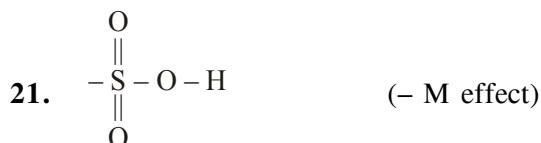
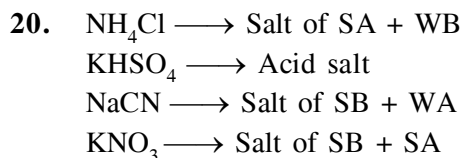
$$\propto \frac{1}{-M, -H, -I}$$





1000 m.eq. $\text{H}_2\text{SO}_4 = 57.1$

30 m.eq. $\text{H}_2\text{SO}_4 = \frac{57.1 \times 30}{1000} = 1.7 \text{ kJ}$



22. Active (Acidic) Hydrogen containing compound liberate CH_4 with $\text{CH}_3 - \text{Mg} - \text{I}$.

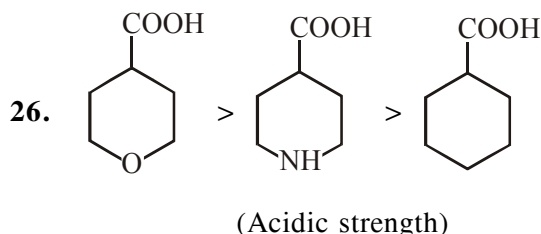
23. $\text{H}_2\text{O} > \text{H}_2\text{O}_2$
strength of H-bond

24.
$$\lambda = \frac{c}{\nu}$$

$$= \frac{3 \times 10^8}{\frac{9000}{3 \times 60}} = \frac{3 \times 10^8 \times 3 \times 60}{9000} = 6 \times 10^6 \text{ m}$$

$$= 6 \times 10^8 \text{ cm}$$

25. $K_{sp} = [\text{Pb}^{++}] [\text{OH}^-]^2$
 $1.2 \times 10^{-5} = 0.12 [\text{OH}^-]^2$
 $10^{-4} = [\text{OH}^-]^2$
 $10^{-2} = [\text{OH}^-]$
 $2 = \text{p}^{\text{OH}}$
 $14 - 2 = \text{pH}$



29.
$$\frac{1}{\lambda} = R \times 3^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

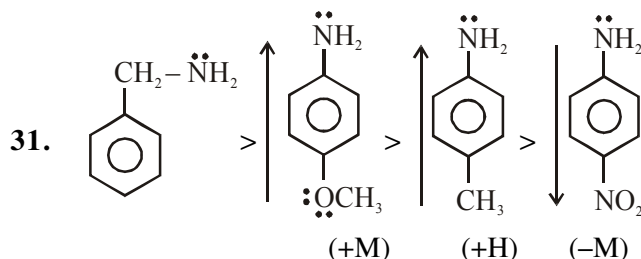
$$\frac{\nu}{c} = R \times 3^2 \left(\frac{4-1}{4^2} \right)$$

$$\nu = \frac{R \times C \times 27}{16}$$

30. $\Delta H - \Delta E = \Delta n_g RT$

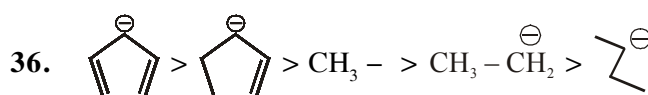
$$= \frac{1}{2} \times 0.002 \times T \text{ Kcal}$$

$$= T \text{ cal}$$

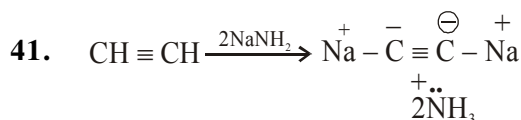


34. $\text{Cu}^+ : 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}$
 $m \quad 0 \quad 0 \quad 1, 0, -1, 0, +1 \quad -2 \quad -1, 0, 1, 2$
 $m = 0, \text{ no. } e^- = 6 \times 2 = 12$
 $m = 1, \text{ no. of } e^- = 3 \times 2 = 6$

35. $\Delta E = q + w$
 $q = 0, \text{ adiabatic process}$
 $\Delta E = W$



39. Moles of $\text{H}_2 = \frac{15}{22.4}$ (maximum)



44.
$$M_{\min} = \frac{\text{element atomic weight} \times 100}{\% \text{ of element}}$$

$$= \frac{32 \times 100}{34} = 94.11$$

46. NCERT Pg.#47
 47. NCERT Pg.#102
 49. NCERT Pg.#147
 52. NCERT-XI, Pg#16,18
 53. NCERT-XI, Pg#87, Para-6.1.2.2
 54. NCERT-XI, Pg#259
 55. NCERT-XI, Pg#295,296
 56. NCERT Pg.#51
 57. NCERT Pg.#101
 59. NCERT Pg.#125, 126
 62. NCERT-XI, Pg#201, Module Pg#19
 63. NCERT-XI, Pg#93, Para-6.3.5
 64. NCERT-XI, Pg#263

65. NCERT-XI, Pg#297
66. NCERT XI Pg.#52
67. NCERT Pg.#114
69. NCERT Pg.#128
72. NCERT-XI, Pg#22,23,24
73. NCERT-XI, Pg#96, Para-6.4.1.4
74. NCERT-XI, Pg#259
75. NCERT-XI, Pg#299
76. NCERT XI Pg.#52
77. NCERT Pg.#113
79. NCERT Pg.#131
82. Module Pg#135
83. NCERT-XI, Pg#71, Para-5.3.3
84. NCERT-XI, Pg#271
85. NCERT-XI, Pg#259
86. NCERT XI Pg.#55
89. NCERT Pg.#134
93. NCERT-XI, Pg#71, Para-5.4
94. NCERT-XI, Pg#271,272
95. NCERT-XI, Pg#275
96. NCERT XI Pg.#59,60
Aptenodytes (Penguin) is flightless birds
98. NCERT Pg.#146
99. NCERT Pg.#133,134
101. NCERT-XI, Pg#1,4,5
Consciousness is character of only living
102. NCERT Pg#37
103. NCERT-XI, Pg#74, Para-5.5.1.2
104. NCERT-XI, Pg#275
105. NCERT-XI, Pg#286
106. NCERT XIth Pg.#57
108. NCERT Pg.#146
109. NCERT Pg.#136
111. NCERT-XI, Pg#7
113. NCERT-XI, Pg#76, Para-5.6
114. NCERT-XI, Pg#279-280
116. NCERT XIth Pg.#22
118. NCERT Pg.#148
119. NCERT Pg.#140
121. NCERT-XI, Pg#9, para of species
122. NCERT-XI, Pg#85, Para-6.1.1
124. NCERT-XI, Pg#287
126. NCERT Pg.#103
128. NCERT Pg.#149
131. NCERT-XI, Pg#2,8,10,11,13
132. NCERT-XI, Pg#86, Para-6.1.2.1
134. NCERT-XI, Pg#287

1001CMD303119064

136. For maximum/minimum value

$$\frac{dy}{dx} = 0 \Rightarrow 5(2x) - 2(1) + 0 = 0 \Rightarrow x = \frac{1}{5}.$$

Now at $x = \frac{1}{5}$, $\frac{d^2y}{dx^2} = 10$ which is positive so y

has minimum value at $x = \frac{1}{5}$.

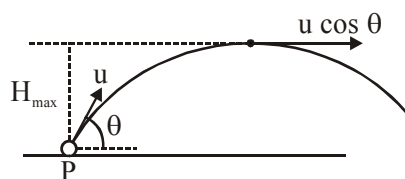
$$\text{Therefore } y_{\min} = 5\left(\frac{1}{5}\right)^2 - 2\left(\frac{1}{5}\right) + 1 = \frac{4}{5}$$

137. $a = g \tan \theta$

$$\Rightarrow \frac{F}{m_1 + m_2} = \frac{QE}{m_1 + m_2} = g \tan \theta$$

$$\therefore E = [m_1 + m_2] \frac{g \tan \theta}{Q}$$

- 138.



Angular momentum at point P

$$L = (mu \cos \theta) H_{\max}$$

$$= mu \cos \theta \frac{u^2 \sin^2 \theta}{2g}$$

$$L = \frac{mu^3 \sin^2 \theta \cos \theta}{2g}$$

139. $\Delta L = L \propto \Delta T$

$$\text{slope} \Rightarrow \frac{\Delta L}{\Delta T} = L \alpha$$

$$(\text{Slope})_A < (\text{Slope})_B ; L_A \alpha_A < L_B \alpha_B$$

$$\therefore L_A = L_B \text{ therefore } \therefore (\alpha_A < \alpha_B)$$

140. $a = -200x + 50$

$$a = -200 \left[x - \frac{50}{200} \right]$$

$$a = -200X' \quad \text{When } X' = x - \frac{50}{200}$$

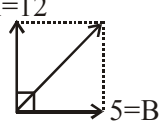
This function is a SHM, periodic & oscillatory also.

$$141. \left. \begin{array}{l} A + B = 17 \\ A - B = 7 \end{array} \right\} \begin{array}{l} \text{Min.} \\ A - B \leq R \leq A + B \\ \text{Max.} \end{array}$$

$$2A = 24$$

$$A = 12$$

$$B = 5$$

$$A=12$$


$$|\vec{R}| = |\vec{A} + \vec{B}| = \sqrt{5^2 + 12^2} = 13$$

$$142. w_{\text{All}} = \Delta k$$

$$w_G + w_f = 0$$

$$mg[5 - h'] - \mu mg \ell = 0$$

$$mg[5 - h'] = \mu mg \ell$$

$$[5 - h'] = 0.5 \times 8$$

$$h' = 1 \text{ m}$$

$$143. v = \sqrt{\frac{2gssin\theta}{1 + \frac{K^2}{R^2}}} = \sqrt{50}$$

$$144. m s \Delta \theta = m' L$$

$$80 \times 1 \times (30 - 0) = m' \times 80$$

$$m' = 30 \text{ g}$$

$$145. V = \omega \sqrt{a^2 - \left(\frac{2a}{3}\right)^2}$$

$$V = \omega \sqrt{\frac{5}{9}a^2}$$

$$V = \frac{\sqrt{5}}{3} a\omega$$

so the new amplitude is given by

$$V_{\text{new}} = \omega \sqrt{a_{\text{New}}^2 - x^2}$$

$$\sqrt{5}a\omega = \omega \sqrt{a_{\text{New}}^2 - \frac{4a^2}{9}}$$

$$5a^2 = a_{\text{New}}^2 - \frac{4a^2}{9}$$

$$a_{\text{New}}^2 = a^2 \left[5 + \frac{4}{9} \right]$$

$$a_{\text{New}} = a \sqrt{\frac{49}{9}} = \left(\frac{7a}{3} \right)$$

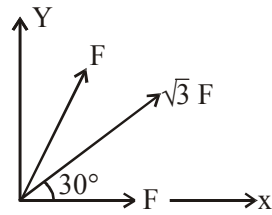
146. By putting the dimensions of each quantity on both the sides we get $[T^{-1}] = [M]^x [MT^{-2}]^y$

Now comparing the dimensions of quantities in both sides we get $x + y = 0$ and $2y = 1$

$$\therefore x = \frac{1}{2}, y = \frac{1}{2}$$

$$147. P = \frac{dW}{dt}$$

$$W = \int dW = \int P dt = \int_1^3 (3t^2 - 2t) dt = \Delta \text{ K.E.}$$

$$148.$$


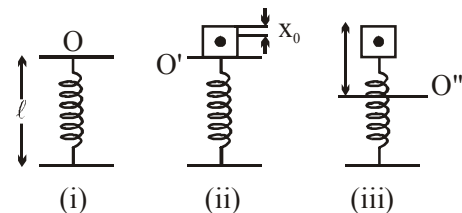
$$F_R = (\sqrt{3}) \left(\frac{GM^2}{a^2} \right) \left(\frac{\sqrt{3}\hat{i} + \hat{j}}{2} \right)$$

$$149. \text{Rate of cooling } R_f = \frac{e_r \sigma A (T^4 - T_0^4)}{ms}$$

$$R_f \propto \frac{A}{m} \propto \frac{A}{V} \propto \frac{1}{r}$$

$$\frac{R_{f_1}}{R_{f_2}} = \frac{r_2}{r_1} = \frac{1}{2} \Rightarrow R_{f_2} = 2R_{f_1}$$

150. The spring has a length l . When m is placed over it, the equilibrium position becomes O' .



If it is pressed from O' (the equilibrium position) to O'' , $O'O''$ is the amplitude

$$OO' = \frac{mg}{k} = \frac{2 \times 10}{200} = 0.10 \text{ m}$$

$$mg = kxO$$

If the restoring force $mA\omega^2 > mg$, then the mass will move up with acceleration, detached from the pan.

i.e., the minimum is just greater than 10 cm. (The actual compression will include x_0 also. But when talking of amplitude, it is always from the equilibrium position with respect to which the mass is oscillating)

151. $\text{Area} = \pi r^2 = \frac{22}{7} \times (2.12)^2 = \frac{98.8768}{7}$
 $= 14.125257 = 14.1 \text{ cm}^2$
 (after rounding off to 3 significant figures as in given data)

154. $\frac{dT}{dt} = \frac{\sigma e A (T^4 - T_0^4)}{ms}$
 $s_A > s_B$ [Other parameter are same]

So $\left(\frac{dT}{dt}\right)_A < \left(\frac{dT}{dt}\right)_B$

155. $\frac{1}{2} K A^2 \sin^2 \omega t = \frac{1}{2} \left(\frac{K A^2}{2} \right) \Rightarrow t = \frac{T}{8}$

156. $a = \frac{dv}{dt} = 6t + 4$

$\int_0^v dv = \int_0^t (16t + 4) dt$

$v = \frac{6t^2}{2} + 4t$

$\int_0^s ds = \int_0^3 \left(\frac{6t^2}{2} + 4t \right) dt$

$s = (t^3 + 2t^2)_0^3$
 $s = (3)^3 + 2(3)^2 = 45\text{m}$

157. Mechanical Energy conservation

$\frac{1}{2} kx^2 = \frac{1}{2} m(5rg)$

$\frac{mg}{2R} x^2 = 5mgR$

$x = \sqrt{10} R$

158. $E_{es} = 2K_0$

159. $C_{V_{\text{mix}}} = \frac{\mu_1 C_{V_1} + \mu_2 C_{V_2}}{\mu_1 + \mu_2} = \frac{1 \times \frac{3}{2} R + 3 \times \frac{5}{2} R}{1 + 3}$

$\frac{18}{8} R = \frac{9}{4} R = \frac{9}{4} \times 8.3 = 18.7$

160. $AB = \lambda$
 $CD = T$

$\frac{AB}{CD} = \frac{\lambda}{T} = v$ (wave speed)

161. $R = \frac{u^2 \sin 2\theta}{g}$ and $H = \frac{u^2 \sin^2 \theta}{2g}$

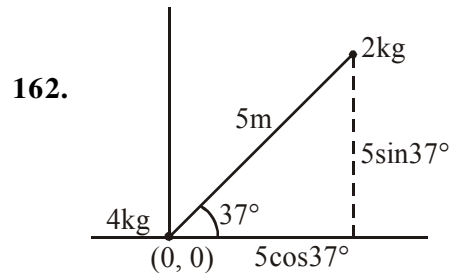
But, $R = 4\sqrt{3}H$

$\frac{u^2 \sin 2\theta}{g} = 4\sqrt{3} \left(\frac{u^2 \sin^2 \theta}{2g} \right)$

$\sin 2\theta = 4\sqrt{3} \frac{\sin^2 \theta}{2}$

$2 \sin \theta \cos \theta = \frac{4\sqrt{3} \sin^2 \theta}{2}$

$\frac{1}{\sqrt{3}} = \frac{\sin \theta}{\cos \theta} = \tan \theta$ or $\theta = 30^\circ$



4kg (0, 0)

2kg (4, 3)

$x_{\text{cm}} = \frac{4(0) + 2(4)}{4 + 2} = \frac{4}{3}$

$y_{\text{cm}} = \frac{4(0) + 2(3)}{4 + 2} = 1$

163. $U = \frac{1}{2} \frac{AY}{L} (\Delta L)^2$

$\frac{U}{2L} = u = \frac{1}{2} Y \left(\frac{\Delta L}{L} \right)^2$

$\frac{u_1}{u_2} \left[\frac{(\Delta L)_1}{(\Delta L)_2} \right]^2 = \left(\frac{A_2}{A_1} \right)^2 = \left[\frac{\frac{\pi}{4} (d)^2}{\frac{\pi}{4} (2d)^2} \right]^2 = \frac{1}{16}$

164. $\Delta E_{\text{int}} = 0$ for a complete cycle and for given cycle work done is negative, so from first law of thermodynamics Q will be negative i.e. $Q < 0$.

165. Maximum velocity of the cork

$$= \omega A = 2\pi n A = 2\pi \left(\frac{v}{\lambda} \right) A$$

$$= 2\pi \times \frac{0.3 \times 100}{1.5\pi} \times 0.5 = 20 \text{ cm/sec}$$

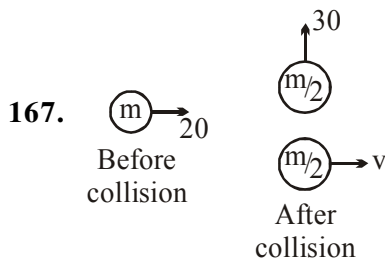
166. Time to cross the river $t = \frac{d}{v_p}$ and drift.

$$x = v_R \times t$$

$$\text{So } x = v_R \frac{d}{v_p}$$

$$300 = 3 \text{ km/hr} \times \frac{500}{v_p}$$

$$v_p = 5 \text{ km/hr}$$



$$\vec{P}_i = \vec{P}_f$$

$$m \times 20\hat{i} = \frac{m}{2} \times 30\hat{j} + \frac{m}{2} \times \vec{v}$$

$$\vec{v} = 40\hat{i} - 30\hat{j}$$

$$|\vec{v}| = 50$$

168. $A_1 v_1 = A_2 v_2$

$$\Rightarrow \frac{v_2}{v_1} = \left(\frac{r_1}{r_2} \right)^2 \Rightarrow r_2 = 2r_1$$

$$\Rightarrow v_2 = v_1 \left(\frac{r_1}{2r_1} \right)^2 \Rightarrow \frac{v_1}{4} = \frac{v}{4}$$

169. $(\Delta Q)_p = \mu C_p \Delta T \Rightarrow 207 = 1 \times C_p \times 10$

$$\Rightarrow C_p = 20.7 \frac{\text{Joule}}{\text{mol-K}}. \text{ Also } C_p - C_v = R$$

$$\Rightarrow C_v = C_p - R = 20.7 - 8.3 = 12.4 \frac{\text{Joule}}{\text{mole-K}}$$

$$\text{So, } (\Delta Q)_v = \mu C_v \Delta T = 1 \times 12.4 \times 10 = 124 \text{ J}$$

170. $1 \text{ b/s} = \frac{1}{1}$

$$1 \text{ b/s} = \frac{1}{1}$$

$$2 \text{ b/s} = \frac{1}{2} \times \frac{2}{2} = 2 \text{ b/sec}$$

171. $F = \frac{2\mu s \sin \theta}{t}$

$$F = \frac{2 \times 0.1 \times 10 \times \frac{1}{2}}{0.1} = 10 \text{ N}$$

172. $I = Mr^2$

Torque acting on the cylinder by pulling the rope with a force F is

$$\begin{aligned} \tau &= rF \sin \theta \\ &= F \cdot (r \sin \theta) = F \cdot r_1 \\ &= 80 \times 0.5 = 40 \text{ N-m} \end{aligned}$$

$$\text{and } I = Mr^2 = 2 \times (0.5)^2 = 0.5 \text{ kg-m}^2$$

$$\text{Now } \tau = I\alpha$$

$$\therefore \alpha = \frac{\tau}{I} = \frac{40 \text{ N-m}}{0.5 \text{ kg-m}^2} = 80 \text{ rad/s}^2$$

173. The meniscus will adjust itself so that the water does not spill.

174. $\eta_1 = 1 - \frac{T_L}{T_H} = \frac{W}{Q_1} = \frac{1}{6}$

$$\text{or } 5T_H - 6T_L = 0 \quad \dots(1)$$

$$\eta_2 = 1 - \frac{T_L - 62}{T_H} = 2\eta_1 = \frac{1}{3}$$

$$\text{or } 2T_H - 3T_L = -186 \quad \dots(2)$$

$$\therefore T_H = 372 \text{ K} = 99^\circ\text{C}$$

$$T_L = \frac{5}{6} T_H = \frac{5}{6} \times 372 \text{ K} = 310 \text{ K} = 37^\circ\text{C}$$

175. When listener is moving towards the sources,

$$v' = \frac{v + v_0}{v} v \quad \text{or} \quad 200 = \frac{v + 40}{v} v \quad \dots(i)$$

where v' is apparent frequency and v is original frequency of source v is velocity of sound in air.

When listener is moving away from the source,

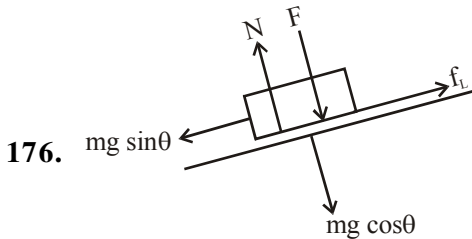
$$v'' = \frac{v - v_0}{v} v \quad \text{or} \quad 160 = \frac{v - 40}{v} v \quad \dots(ii)$$

From equations (i) and (ii)

$$\frac{200}{160} = \frac{v + 40}{v - 40}$$

Solving, we get ;

$$v = 360 \text{ m/s.}$$



$$f_L = mg \sin \theta$$

$$\mu N = mg \sin \theta$$

$$\mu(F + mg \cos \theta) = mg \sin \theta$$

$$F = \frac{mg}{\mu} (\sin \theta - \mu \cos \theta)$$

177. $2\pi R = L$

$$R = \frac{L}{2\pi}$$

$$I = MR^2 = M \left(\frac{L}{2\pi} \right)^2$$

$$= \frac{ML^2}{4\pi^2}$$

178. $I_1 \omega_1 = I_2 \omega_2$

$$\frac{1}{2} MR^2 \omega = \left(\frac{1}{2} MR^2 + \frac{1}{2} \left(\frac{M}{4} \right) R^2 \right) \omega_2$$

$$\omega_2 = \frac{4}{5} \omega$$

179. Coefficient of performance $K = \frac{T_2}{T_1 - T_2}$

$$= \frac{(273 - 23)}{(273 + 27) - (273 - 23)} = \frac{250}{300 - 250} = \frac{250}{50} = 5$$

180. First overtone of string A = Second overtone of string B.

$\Rightarrow$ Second harmonic of A = Third harmonic of B

$$\Rightarrow n_2 = n_3 \Rightarrow [2(n_1)]_A = [3(n_1)]_B$$

$$(\because n_1 = \frac{1}{2\ell} \sqrt{\frac{T}{\pi r^2 \rho}})$$

$$\Rightarrow 2 \left[\frac{1}{2\ell_A r_A} \sqrt{\frac{T}{\pi \rho}} \right] = 3 \left[\frac{1}{2\ell_B r_B} \sqrt{\frac{T}{\pi \rho}} \right]$$

$$\frac{\ell_A}{\ell_B} = \frac{2}{3} \frac{r_B}{r_A} \Rightarrow \frac{\ell_A}{\ell_B} = \frac{2}{3} \times \frac{r_B}{(2r_B)} = \frac{1}{3}$$